

Letter of Collaboration | External FreeMoCap Users and Contributors

31 August 2026

To the National Science Foundation,

This document compiles brief statements from external FreeMoCap users and contributors who have independently become involved with the project through animation, software development, interface design, and professional game production. Together, these statements describe their existing contributions and use of FreeMoCap, as well as their anticipated continued involvement in the FreeMoCap ecosystem.

Andres Casals

Animation / Blender

My name is Andres Casals, and I am a Telecommunications Engineer. I first became involved with FreeMoCap through a personal animation project, creating a music video in Unreal Engine. When I reached the character animation stage, I tried several free tiers of commercial motion capture software, but none produced usable results. After searching YouTube for open-source alternatives, I stumbled upon the FreeMoCap project. It not only provided the tools I needed for my project, but also gave me complete freedom to modify the software and contribute back to the community.

Initially, my contributions focused on the Blender animation export pipeline to support my own workflow goals: correcting marker orientation, applying rigid body logic to markers, generating an armature compatible with the Unreal Engine MetaHuman, implementing a foot-locking function, and improving the Blender add-on UI to streamline exporting animations to Unreal Engine.

Given my engineering background, the scientific aspects of FreeMoCap also naturally appealed to me. I contributed several visualization tools in Blender, including Base of Support, joint angles, and overlay charts for joint angles and COM position. Additionally, I have contributed to user onboarding and instructional materials by creating video tutorials, live demos, camera benchmarks, and testing new features.

I look forward to continuing my work with FreeMoCap, primarily focusing on the animation pipeline—an area I am passionate about. I strongly align with the open-source ethos of making motion capture tools as accessible as possible. My goal is to keep improving the software with user-friendly, animation-focused tools and well-documented workflows.

Andres Casals · Relationship to proposing team: External FreeMoCap contributor

Dominic Scott

Pose Estimation / Real-Time Analysis

I work with realtime motion capture tools, with a focus on making open source tools easier to deploy and use in real-world applications. Previously I created a motion capture plugin for a realtime graphics tool called TouchDesigner, and became involved in FreeMoCap when I saw the work being done to create a realtime pipeline as well. I wanted to contribute the knowledge I had learned from previous projects to help, and have since contributed across several parts of the technical pipeline.

My work with FreeMoCap has focused particularly on making pose-estimation models easier to deploy and on developing capabilities for real-time metrics and analysis. This has included specifying a standard for model metadata that allows for drag-and-drop inclusion of new models into FreeMoCap, and acceleration techniques to run models as quickly as possible across all platforms and hardware types. A recurring goal of this work has been reducing the technical overhead required to bring new models into practical use within FreeMoCap and expanding cross-platform compatibility.

Going forward, I hope to contribute further to the development of FreeMoCap's pose-estimation and real-time analysis capabilities. I am particularly interested in continuing to improve how new models can be integrated into the software and in helping develop workflows that make these capabilities easier to extend and use.

Dominic Scott · Relationship to proposing team: External FreeMoCap contributor

Pooya Moradi Motlagh

Product Design / UI/UX / Frontend Development

My name is Pooya Moradi M, and my background is in product design, UI/UX, and frontend development. I became involved with FreeMoCap as an external contributor.

My contributions have mainly focused on the user interface and overall usability of the software. This has included designing interfaces and user experiences in Figma, building frontend components using React and CSS, and creating a shared design system that provides a consistent foundation for the project and makes it easier for me and other contributors to work on the interface. A lot of this work has been about making the software feel clearer, more intuitive, and easier to use.

I plan to continue contributing to FreeMoCap as the software evolves, particularly through UI/UX design, frontend development, improving the design system, and making user-facing workflows easier to understand and use.

Pooya Moradi Motlagh · Rome, Italy · Product Designer · pooyamoradim@gmail.com · GitHub: @pooyadeperson 28 Aug 2026 · Relationship to proposing team: External FreeMoCap contributor

Jonathan Nilsson — GIBBET GAMES AB

Professional Game Development / Animation

I am writing on behalf of GIBBET GAMES AB regarding our use of and collaboration with the FreeMoCap project. GIBBET GAMES is an independent video game studio in Skövde, Sweden, working with game development, animation, rigging, motion capture, and technical animation.

We began using FreeMoCap during our work on *Esoteric Ebb*, where we used markerless motion capture extensively for character animation. FreeMoCap allowed us to record motion directly in our own workspace and iterate much faster than when relying on access to a dedicated OptiTrack studio.

We have since continued developing FreeMoCap-based workflows for game production. This includes transferring motion data through Blender into Autodesk Maya and Advanced Skeleton, as well as developing our own Maya tools for working with FreeMoCap animation, synchronized camera views, and recorded video.

Through our use of FreeMoCap, we have been in contact with members of the development team and have provided feedback from the perspective of game and animation production. We plan to continue using FreeMoCap in future projects and are interested in continuing to test, develop, and share animation-focused workflows with the FreeMoCap community.

If the proposed PESOSE project is funded, we would be happy to remain involved as an external user and collaborator and provide feedback from professional game-development and animation workflows.

Jonathan Nilsson · GIBBET GAMES AB · Relationship to proposing team: External FreeMoCap user and collaborator